UNIVERSITÄT
LEIPZIG

Asymmetric Numeral System

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Outline

Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

1st frame

- This is one of the section[1].
- Let's check for 2nd footnote [2].
- The footnotes will be added from the bottom. If you want to pull the footnotes towards red-black margin line then use `\addfootspace{}` at the end of the `\frame{}`.
- `\addfootspace{}` actually adds an additional vacant footnote (without counting that vacant footnote).
- To add more foot space add more `\addfootspace{}`.

[1] This is a long footnote. Here I am. Who am I?

[2] This is a 2nd long footnote. Here I am. Who am I?

2nd frame

- Here is the next section[3]
- Here I used two `\addfootspace{}`. Don't do `\addfootspace{\vspace{1cm}}`.

[3] This is a 2nd long footnote. Here I am. Who am I?

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding [3]

Basic Idea

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  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}

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- Symmetric or Asymmetric
- rANS and tANS

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- $\bar{s} : \mathbb{N} \to \mathcal{A}, \bar{s}(x) = \text{mod}(x, b)$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
  e.g. splits natural numbers into subsets of even/odd numbers [1]

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

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- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$X_{out} = 202518$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$
\begin{aligned}
X_{\text{init}} &= 0 \\
X_1 &= X_{\text{init}} \times 10 + 2 = 2 \\
X_2 &= X_1 \times 10 + 0 = 20 \\
X_3 &= X_2 \times 10 + 2 = 202 \\
X_4 &= X_3 \times 10 + 5 = 2025 \\
X_5 &= X_4 \times 10 + 1 = 20251 \\
X_{\text{out}} &= X_5 \times 10 + 8 = 202518
\end{aligned}
$$

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$$X_{i+1} = X_i \times 10 + s_i$$

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- x-th appearance of symbol s

$$X_{i+1} = C(s_i, X_{i+1}) = X_i \cdot 10 + s_i$$

Symmetric or Asymmetric

rANS

- Here is the next section[4]
- Here I used two `\addfootspace{}`. Don't do `\addfootspace{\vspace{1cm}}`.

[4] This is a 2nd long footnote. Here I am. Who am I?

tANS

Performance

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

1. J. Duda, K. Tahboub, N. J. Gadgil and E. J. Delp, ”The use of asymmetric numeral systems as an accurate replacement for Huffman coding,” 2015 Picture Coding Symposium (PCS), Cairns, QLD, Australia, 2015, pp. 65-69, doi: 10.1109/PCS.2015.7170048.
2. 2
3. https://en.wikipedia.org/wiki/Asymmetric_numeral_systems